

- 29** Light of wavelength  $567\text{ nm}$  is incident normally on a diffraction grating. The grating has 400 lines per mm. A number of diffraction maxima are observed on the far side of the grating.

What is the angle between the second-order maximum and the third-order maximum?

- A**  $13.1^\circ$                       **B**  $13.9^\circ$                       **C**  $15.9^\circ$                       **D**  $27.0^\circ$

- 20** Two cylindrical conductors,  $X$  and  $Y$ , are made from the same material. The conductors have